

(10 pts) 1. Explain the difference between a community, an assemblage, and an ecosystem.

(10 pts) 2. Explain the difference between ecosystem function and ecosystem services and the relationship between them. Provide an example of your choosing to show this relationship.

(5 pts) 3. Why is a population-based approach to ecosystem restoration often ineffective?

(15 pts) 4. Name the other three non-population-based approaches to ecosystem restoration and explain the difference in emphasis between them. Explain why that emphasis is important for *each* approach.

(10 pts) 5. Diversity measures like H' incorporate richness and evenness. Explain one disadvantage of these diversity measures from the perspective of ecosystem function. Explain a more suitable way of measuring diversity and why it is suitable for assessing ecosystem function.

(5 pts) 6. Explain how complementarity increases ecosystem function.

(10 pts) 7. Explain the difference between the insurance hypothesis and the portfolio effect. Explain how these confer stability under the high redundancy model of ecosystem function.

(5 pts) 8. A comprehensive ecosystem-based management plan must account for all sectors used by humans. Explain why this is important in terms of trade-offs among sectors.

(5 pts) 9. Which do you think provides greater economic value to humans, a shrimp farm or a mangrove forest? Justify your answer.